

VITTORIO CENTORE

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Education

University of Michigan

B.S. in Computer Science

Ann Arbor, MI

Expected May 2028

- **GPA:** 3.72/4.00
- **Coursework:** Data Structures & Algorithms, Web Systems, Object Oriented Programming, Computer Organization, Theory of Computer Science, Discrete Mathematics, Linear Algebra, Intro to Data Science
- **Activities:** Atlas Digital Tech Consulting, Reach Consulting Group, Michigan Rover Team

Experience

Capital One

Incoming Software Engineering Intern

Chicago, IL

Jun 2026 - Aug 2026

Clear Estimates

Software Developer

Ann Arbor, MI

Jan 2026 – Present

- Trained a PyTorch autoencoder on 903k+ historical construction plan estimates to predict missing line items from partial project inputs, achieving 66.8% Recall@10 and outperforming an ALS baseline by 9.3%
- Deployed model into Clear Estimate's React web platform via Docker and ASP.NET microservices, enabling real-time estimate recommendations for 10.7k+ monthly active projects and saving \$112k/month

AgentMail (YC S25)

Software Engineering Analyst

Ann Arbor, MI

Sep 2025 – Dec 2025

- Developed a full-stack, interactive demo on the company website that visualized real API calls through a storytelling experience, improving developer trial conversion rates and boosting product engagement by a theoretical 16%.
- Architected a concurrent backend pipeline using FastAPI background tasks and the OpenAI Async SDK (asyncio), offloading latency-heavy API calls to prevent blocking and improving effective throughput under 5s latency.
- Engineered an AI sales agent using OpenAI SDK and AgentMail's Inbox API reducing manual workload by 64%

'Sup | Seed Round Startup

Software Engineering Intern

Miami, FL

Jun 2025 - Sep 2025

- Developed a mock interview tool with Next.js and FastAPI, integrating Gemini's API to analyze user responses without speech-to-text conversion, enabling faster, more precise feedback and improving practice efficiency by 40%.
- Deployed facial tracking with MediaPipe and face-api.js to derive emotion-based confidence signals, feeding them into an LLM-based grading system to deliver more personalized feedback and improve evaluation accuracy by 59%.
- Constructed a high-performance media compression pipeline with FFmpeg that reduced audio/video storage by 83%

U-M School of Information

Software Engineer

Ann Arbor, MI

Feb 2025 – May 2025

- Designed scalable pipeline that recommends faculty for awards by matching profiles to award requirements and ranking top candidates, powering a React chat interface and processing 8,000 profiles in ~13 seconds
- Optimized recommendation pipeline using Pandas and PyTorch CUDA to cut latency by 32%, then launched updates through CI/CD pipelines with unit + integration tests for faculty ranking logic, improving correctness by 26%

Projects

Limit Order Book Engine

- Implemented a multithreaded C++ limit order book over TCP with lock-free SPSC ring buffers, sustaining 80K+ orders/sec at sub-2 μ s median latency across 10+ concurrent clients, 40 $\times$ faster than a mutex-locked baseline
- Integrated a Python replay harness that streams 5M+ market events to the engine over TCP, benchmarking p50/p99/p999 end-to-end latency across 10 symbols and surfacing a 3 $\times$ tail-latency spike under thread contention

Rush – Student Org Platform

- Built a campus recruiting platform centralizing 1800+ UMich clubs, enabling students to discover orgs via a personalized feed and track application statuses, alongside a recruiter portal for clubs to manage applicant pipelines
- Engineered an ML recommendation pipeline using pgvector for ANN retrieval, LightGBM to re-rank candidates by predicted click probability, and rule-based re-ranking for deadline urgency and feed diversity

Technical Skills

Languages: Python, C++, C, Java, TypeScript, JavaScript, HTML/CSS, SQL

Technologies: React.js, Next.js, Tailwind, FastAPI, AWS, Firebase, PostgreSQL, Git, Docker

Interests: Soccer (Napoli Supporter), Cars, Watches, Snowboarding, Marvel, Travel, Music